

# Plasticity:

## The Anchor Concept of the Neurosciences

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# Plasticity

- **A common understanding of plasticity**

The capacity of the nervous system to change in response to changes within the organism and / or its environment. The changes the nervous system undergoes, reflect its adaptation to novel requirements.

- **No comprehensive definition of plasticity**

Different conceptualisations of plasticity correspond to the insights on the dynamic nature of the nervous system.

- **Coevolution of plasticity and the neurosciences**

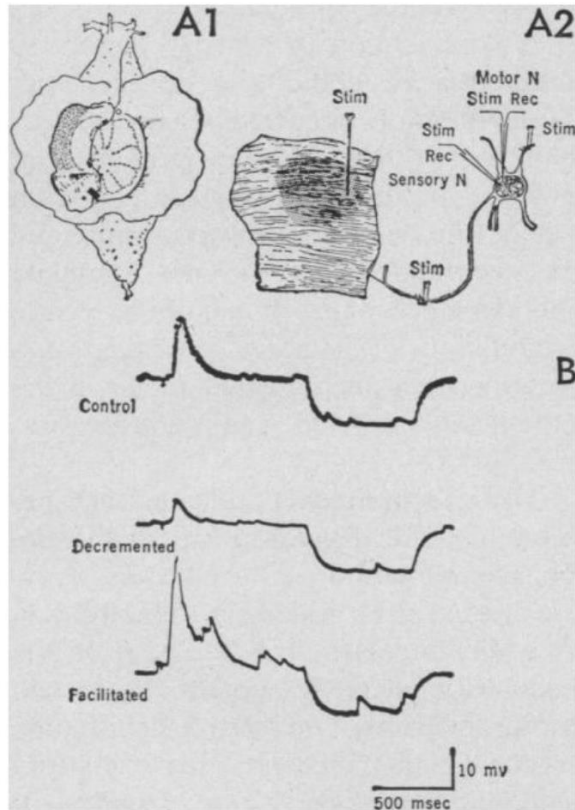
The diversification that plasticity has undergone, reflects the advances made in understanding the changeability of the nervous system.

- **Plasticity changes between different fields within and outside the neurosciences**

The exchange is particularly prominent between neurobiology and psychology, but also comprises other fields, such as philosophy

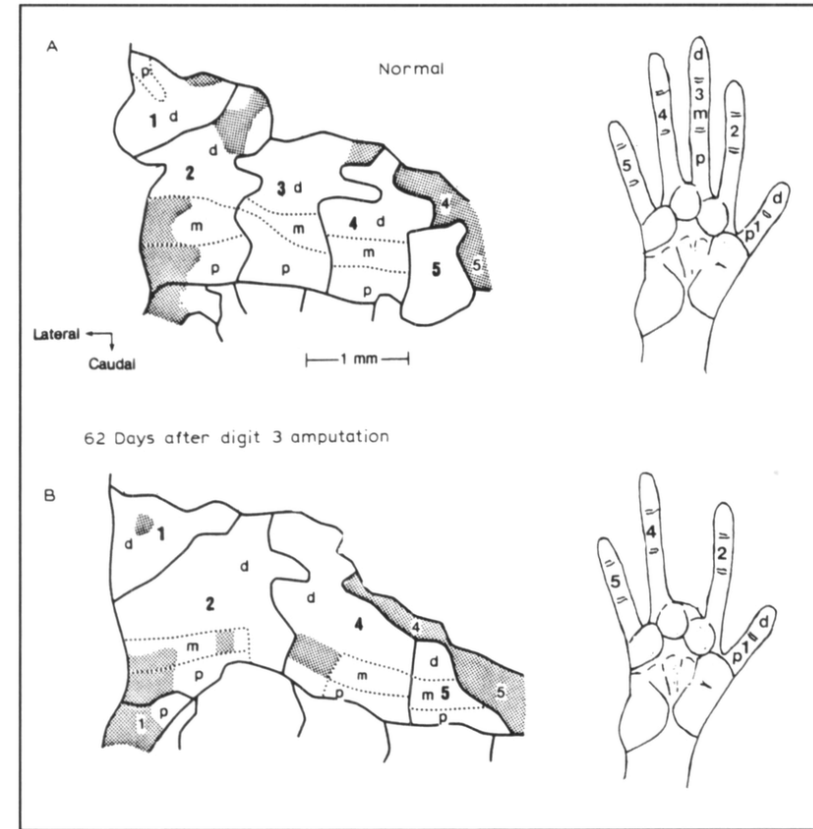
# The diversified use of plasticity

## Synaptic plasticity in the **sea snail aplysia**



"However, the relevance of synaptic plasticity to a specific instance of behavioral modification has never been demonstrated. We have described behavioral parameters of habituation and dishabituation of the gillwithdrawal reflex in the intact *Aplysia* (4), and we have examined their cellular correlates in a semi-intact preparation (5)." (Castelucci et al., 1970)

## Map plasticity in **adult owl monkeys**

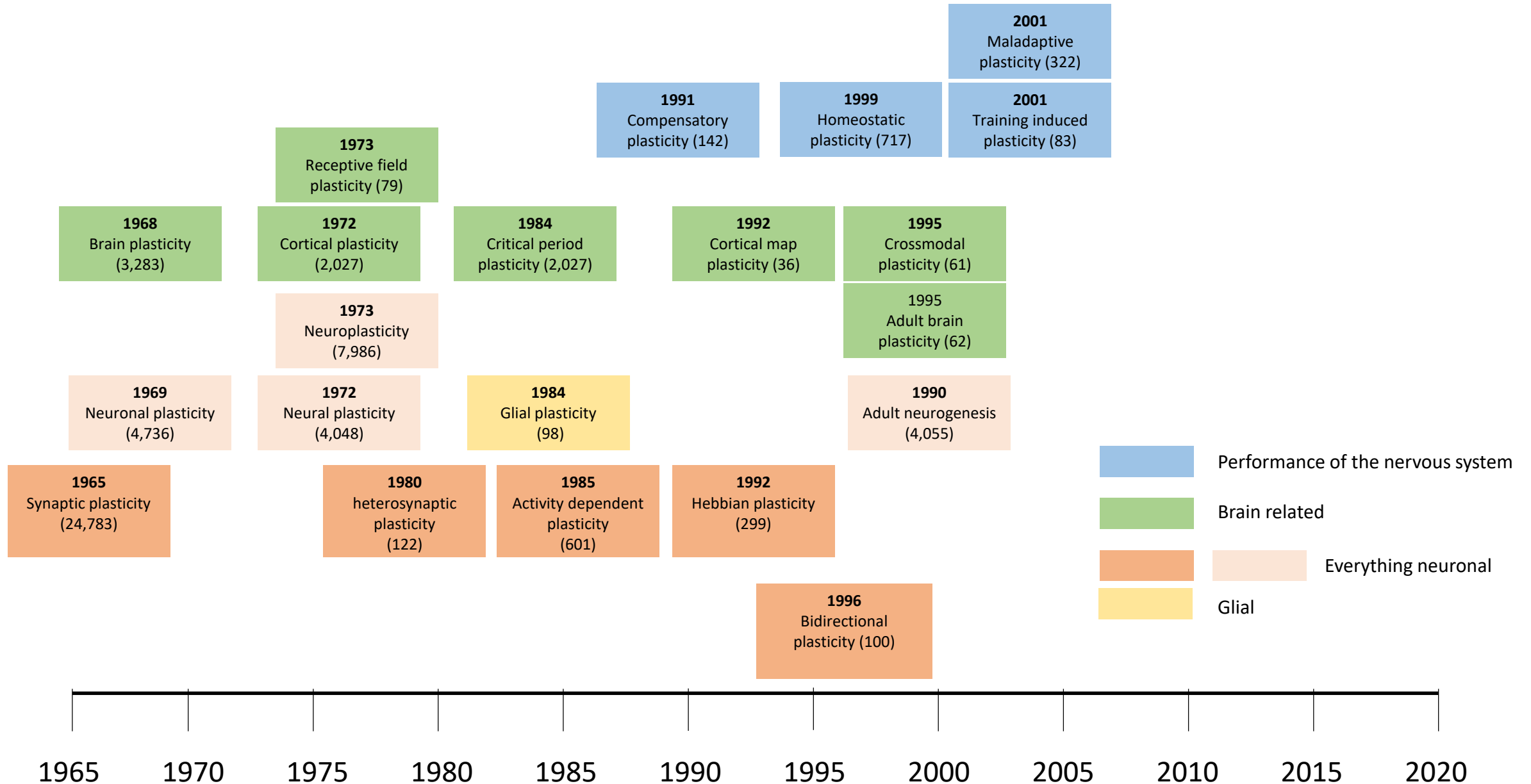


"The present experiments examine the effects of a more severe peripheral lesion, the amputation of one or two digits, extending earlier studies of map "plasticity" in several ways." (Merzenich et al., 1984)

# The evolution of plasticity

- The nervous system becomes plastic: synaptic plasticity (1965)  
Eric Kandel, the conditioning of the sea snail aplysia
- The developing nervous system becomes plastic: cortical plasticity (1972)  
Torsten Wiesel and David Hubel, the visual system of cats
- **The transformation: the adult nervous system becomes plastic: adult neurogenesis (1990), cortical map plasticity (1992)**  
Fernando Nottebohm, changing populations of neurons in songbirds  
Michael Merzenich, changes in cortical maps of apes

# Plasticity concepts in the neuroscience literature



# Plasticity as an anchor concept

**The ontological dimension:** The nervous system is plastic.

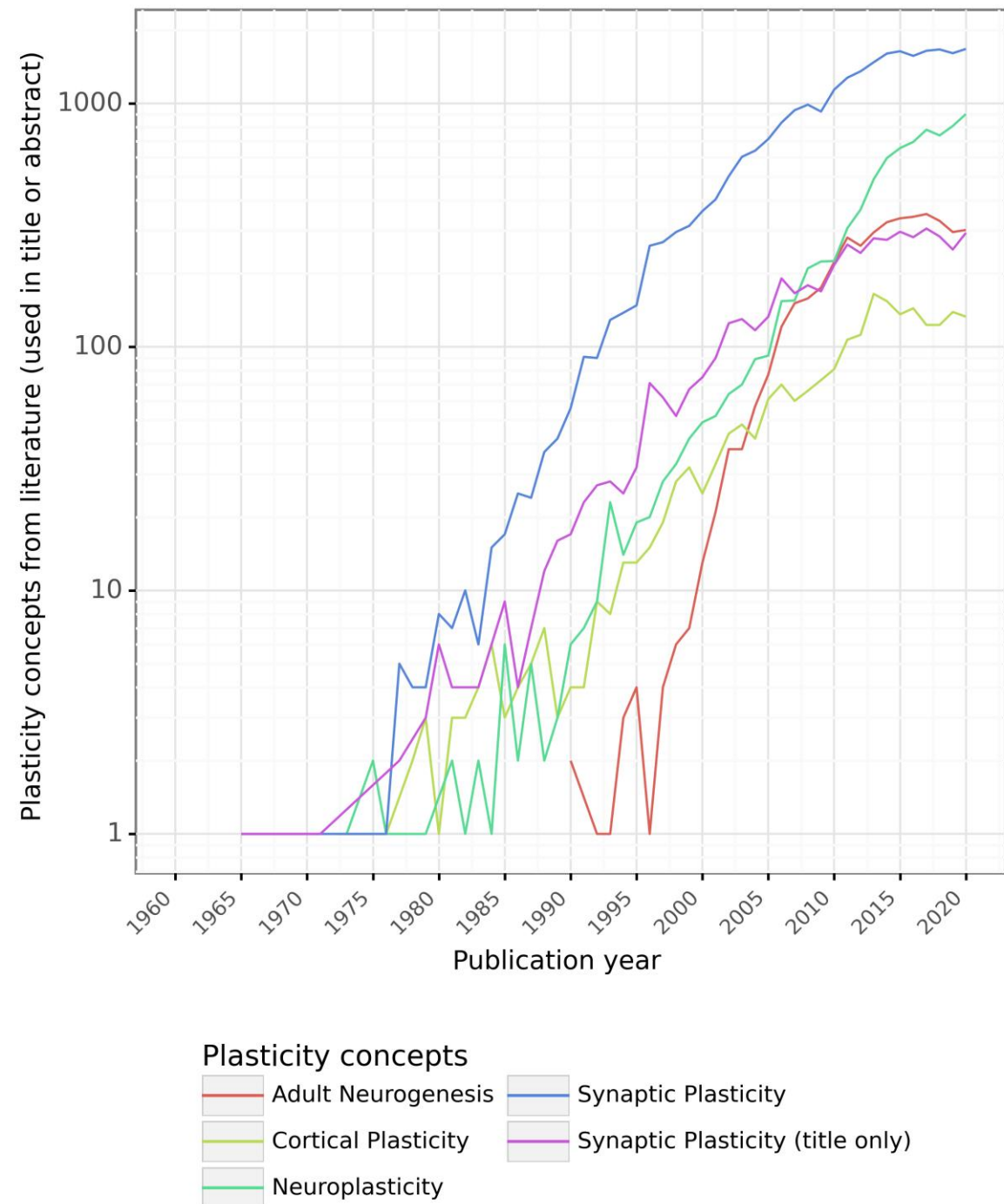
**The epistemological function:** Researchers gather evidence for processes of change and adaptation in the nervous and articulate novel concepts of plasticity. Empirical advances become fixed in word pairs to which other researchers can refer and around which research fields can assemble.

# The use of plasticity

- **Qualitative:** Two series of interviews with pioneers of plasticity (2009 and 2019-2022)
  - Fiona Doetsch (Biozentrum, University of Basel, Switzerland)
  - Michael Merzenich (professor emeritus, University of California, San Francisco)
  - Fernando Nottebohm (professor emeritus, The Rockefeller University, New York)
  - Mriganka Sur (MIT, Boston)
  - Alvar Pascual Leone (Harvard University, Boston)
- **Quantitative:** Bibliometrics of plasticity combining two databases, Pubmed (1965-2020) and Web of Science (1990-2018)

# Dominant plasticity concepts in the literature (PubMed)

- Different conceptualizations of plasticity are differently successful
- Yearly usage of selected keyword combinations searched in “**title**” and “**abstract**” of publications in PubMed
- Earlier and continued dominance of the use of “synaptic plasticity” over other keyword combinations.
- It also confirms that “adult neurogenesis” and “neuroplasticity” has been taken up predominantly after 1990s.
- In addition, the count of publications presented highlights the fact that WOS does not index all publications that are indexed in PubMed.





# Data and methods (1/2)

- Clarivate's Web of Science<sup>1</sup> (WOS), 1990-(end of) 2018, “**article**” and “**review**” publications
- **Corpus construction:** search the lowercased “plasticity” in title, abstract and keywords
- To further limit the usages of this concept to the areas closer (but not limited to) neurosciences, we follow **three** strategies
  - 1) Subject categories by WOS (excluding less relevant fields, but they are covered in more precise strategies)
  - 2) Specific keyword combinations (*word pairs*):
    - 'adult neurogenesis', 'synaptic plasticity', 'cortical plasticity', 'cortical map plasticity', 'receptive field plasticity', 'heterosynaptic plasticity', 'hebbian plasticity', 'structural plasticity', 'neuronal responses', 'plasticity of neuronal responses', 'bidirectional plasticity', 'functional plasticity', 'developmental plasticity', 'critical period plasticity', 'adult brain plasticity', 'ocular dominance plasticity', 'neuronanatomical plasticity', 'cross modal plasticity'

# Data and methods (2/2)

- Two previous approaches could be driven by
  - a) the way WOS defines the subject categories (journal based)
  - b) the keyword combinations highlighted previously in the literature could be driven by the area of focus
- 3) ***A parallel attempt***, we try to apply a more *objective criteria* using text and content analysis (*found words*).
  - **nltk** library in Python (Bird, Loper and Klein, 2009), first tokenize the words used in the title and abstract of publications. We decided to tokenize the text as our first step since we need the punctuation and sentence structure in determining the words. We then remove punctuations from these tokenized text, and lower case all the text elements. We then use plasticity as an “anchor word” and see if it is used once or more in the title and abstract. We identify which are the words pairing with it as being used before it.
  - There were specific cases where the word used before plasticity was not a proper adjective or a scientific term, we excluded those from the following visualizations.

# Example 1 of text analysis using exact word pairs

| PK_ITEMS    | SOURCETITLE                      | UT_EID          | PUBYEAR | DOCTYPE | SC_DESCRIPTION                     | OECD_DESCRIPTION |
|-------------|----------------------------------|-----------------|---------|---------|------------------------------------|------------------|
| 35583791345 | PLOS<br>COMPUTATIONAL<br>BIOLOGY | 000463877900036 | 2019    | Article | BIOCHEMICAL<br>RESEARCH<br>METHODS | NATURAL SCIENCES |

Exact word pairs



Title

Synaptic plasticity onto inhibitory neurons as a mechanism for ocular dominance plasticity<sup>1</sup>

Abstract

Ocular dominance plasticity is a well-documented phenomenon allowing us to study properties of cortical maturation. Understanding this maturation might be an important step towards unravelling how cortical circuits function. However, it is still not fully understood which mechanisms are responsible for the opening and closing of the critical period for ocular dominance and how changes in cortical responsiveness arise after visual deprivation. In this article, we present a theory of ocular dominance plasticity. Following recent experimental work, we propose a framework where a reduction in inhibition is necessary for ocular dominance plasticity in both juvenile and adult animals. In this framework, two ingredients are crucial to observe ocular dominance shifts: a sufficient level of inhibition as well as excitatory-to-inhibitory synaptic plasticity. In our model, the former is responsible for the opening of the critical period, while the latter limits the plasticity in adult animals. Finally, we also provide a possible explanation for the variability in ocular dominance shifts observed in individual neurons and for the counter-intuitive shifts towards the closed eye. Author summary During the development of the brain, visual cortex has a period of increased plasticity. Closing one eye for multiple days during this period can have a profound and life-long impact on neuronal responses. A well-established hypothesis is that the absolute level of inhibition regulates this period. In light of recent experimental results, we suggest an alternative theory. We propose that, in addition to the level of inhibition, synaptic plasticity onto inhibitory neurons is just as crucial. We propose a model which explains many observed phenomena into one single framework. Unlike theories considering only the level of inhibition, we can account for both the onset as well as the closure of this period. Furthermore, we also provide an explanation for the small fraction of neurons that show counter-intuitive behaviour and provide some testable predictions.

<sup>1</sup> I had a typo in ocular in my exact words definition, writing it as "ocular", and this very accident shows how this method is prone to accidental exclusion

# Example 1 of text analysis using anchors

|       | PK_ITEMS              | SOURCETITLE                      | UT_EID               | PUBYEAR                                                                                    | DOCTYPE | SC_DESCRIPTION                     | OECD_DESCRIPTION |
|-------|-----------------------|----------------------------------|----------------------|--------------------------------------------------------------------------------------------|---------|------------------------------------|------------------|
|       | 35583791345           | PLOS<br>COMPUTATIONAL<br>BIOLOGY | 000463877900036      | 2019                                                                                       | Article | BIOCHEMICAL<br>RESEARCH<br>METHODS | NATURAL SCIENCES |
| Title | Word used before<br>↓ | Anchor<br>↓                      | Word used after<br>↓ | Synaptic plasticity onto inhibitory neurons as a mechanism for ocular dominance plasticity |         |                                    |                  |

Abstract

Ocular **dominance plasticity** is a well-documented phenomenon allowing us to study properties of cortical maturation. Understanding this maturation might be an important step towards unravelling how cortical circuits function. However, it is still not fully understood which mechanisms are responsible for the opening and closing of the critical period for ocular dominance and how changes in cortical responsiveness arise after visual deprivation. In this article, we present a theory of ocular **dominance plasticity**. Following recent experimental work, we propose a framework where a reduction in inhibition is necessary for ocular **dominance plasticity** in both juvenile and adult animals. In this framework, two ingredients are crucial to observe ocular dominance shifts: a sufficient level of inhibition as well as excitatory-to-inhibitory **synaptic plasticity**. In our model, the former is responsible for the opening of the critical period, while the latter limits **the plasticity** in adult animals. Finally, we also provide a possible explanation for the variability in ocular dominance shifts observed in individual neurons and for the counter-intuitive shifts towards the closed eye. Author summary During the development of the brain, visual cortex has a period of **increased plasticity**. Closing one eye for multiple days during this period can have a profound and life-long impact on neuronal responses. A well-established hypothesis is that the absolute level of inhibition regulates this period. In light of recent experimental results, we suggest an alternative theory. We propose that, in addition to the level of inhibition, **synaptic plasticity onto** inhibitory neurons is just as crucial. We propose a model which explains many observed phenomena into one single framework. Unlike theories considering only the level of inhibition, we can account for both the onset as well as the closure of this period. Furthermore, we also provide an explanation for the small fraction of neurons that show counter-intuitive behaviour and provide some testable predictions.

# Example 2 of text analysis using exact word pairs

| PK_ITEMS | SOURCETITLE | UT_EID | PUBYEAR | DOCTYPE | SC_DESCRIPTION | OECD_DESCRIPTION |
|----------|-------------|--------|---------|---------|----------------|------------------|
|----------|-------------|--------|---------|---------|----------------|------------------|

Exact word pairs

(None identified)

|         |                          |                 |      |         |               |                                |
|---------|--------------------------|-----------------|------|---------|---------------|--------------------------------|
| 6980386 | NEUROLOGICAL<br>SCIENCES | 000372330000006 | 2016 | Article | NEUROSCIENCES | MEDICAL AND<br>HEALTH SCIENCES |
|---------|--------------------------|-----------------|------|---------|---------------|--------------------------------|

Title

Biological factors and age-dependence of primary motor cortex experimental plasticity

Abstract

To evaluate whether the age-dependence of brain plasticity correlates with the levels of proteins involved in hormone and brain functions we executed a paired associative stimulation (PAS) protocol and blood tests. We measured the PAS-induced plasticity in the primary motor cortex. Blood levels of the brain-derived neurotrophic factor (BDNF), estradiol, the insulin-like growth factor (IGF)-1, the insulin-like growth factor binding protein (IGFBP)-3, progesterone, sex hormone-binding globulin (SHBG), testosterone, and the transforming growth factor beta 1 (TGF-beta 1) were determined in 15 healthy men and 20 healthy women. We observed an age-related reduction of PAS-induced plasticity in females that it is not present in males. In females, PAS-induced plasticity displayed a correlation with testosterone ( $p = 0.006$ ) that became a trend after the adjustment for the age effect ( $p = 0.078$ ). In males, IGF-1 showed a nominally significant correlation with the PAS-induced plasticity ( $p = 0.043$ ). In conclusion, we observed that hormone blood levels (testosterone in females and IGF-1 in males) may be involved in the age-dependence of brain plasticity.

# Example 2 of text analysis using anchors

| PK_ITEMS | SOURCETITLE | UT_EID | PUBYEAR | DOCTYPE | SC_DESCRIPTION | OECD_DESCRIPTION |
|----------|-------------|--------|---------|---------|----------------|------------------|
|----------|-------------|--------|---------|---------|----------------|------------------|

|         |                          |                 |      |         |               |                                |
|---------|--------------------------|-----------------|------|---------|---------------|--------------------------------|
| 6980386 | NEUROLOGICAL<br>SCIENCES | 000372330000006 | 2016 | Article | NEUROSCIENCES | MEDICAL AND<br>HEALTH SCIENCES |
|---------|--------------------------|-----------------|------|---------|---------------|--------------------------------|

Word used before

Anchor

Word used after

Title

Biological factors and age-dependence of primary motor cortex **experimental plasticity**

Abstract

To evaluate whether the age-dependence of **brain plasticity** **correlates** with the levels of proteins involved in hormone and brain functions we executed a paired associative stimulation (PAS) protocol and blood tests. We measured the **PAS-induced plasticity** **in** the primary motor cortex. Blood levels of the brain-derived neurotrophic factor (BDNF), estradiol, the insulin-like growth factor (IGF)-1, the insulin-like growth factor binding protein (IGFBP)-3, progesterone, sex hormone-binding globulin (SHBG), testosterone, and the transforming growth factor beta 1 (TGF-beta 1) were determined in 15 healthy men and 20 healthy women. We observed an age-related reduction of **PAS-induced plasticity** **in** females that it is not present in males. In females, **PAS-induced plasticity** **displayed** a correlation with testosterone ( $p = 0.006$ ) that became a trend after the adjustment for the age effect ( $p = 0.078$ ). In males, IGF-1 showed a nominally significant correlation with the **PAS-induced plasticity** ( $p = 0.043$ ). In conclusion, we observed that hormone blood levels (testosterone in females and IGF-1 in males) may be involved in the age-dependence of **brain plasticity**.

# Plasticity and its use in different scientific fields

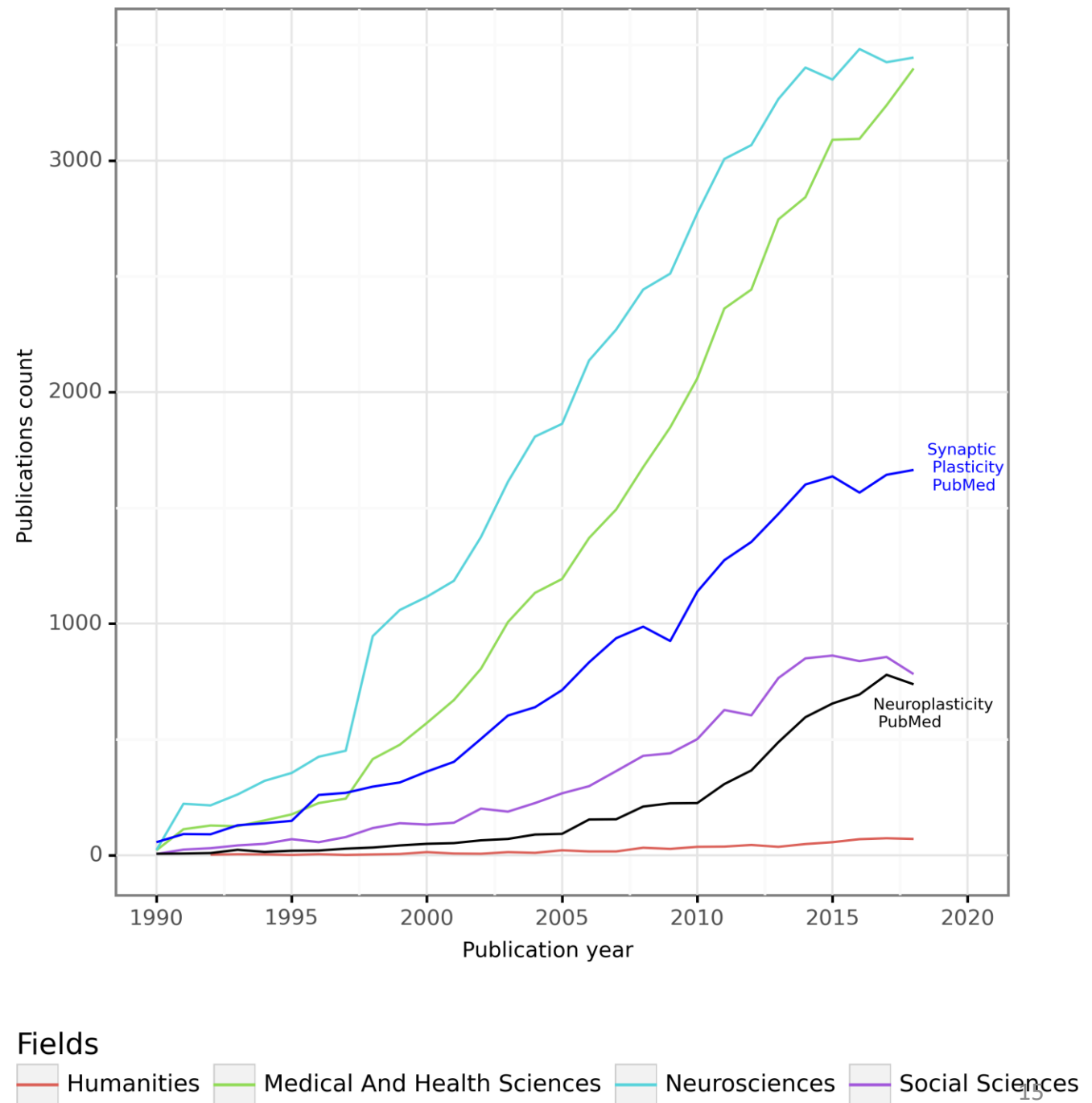
A total of 101,198 publications once we exclude fields that are not directly relevant for the neurosciences and plasticity.

These include 653 publications in Humanities, 38,999 in Medical and Health Sciences, 51,594 in Neurosciences and 9,952 in Social Sciences that use plasticity

Exact usage of “**plasticity**” in **title, abstract** or **keywords** of publications in selected fields

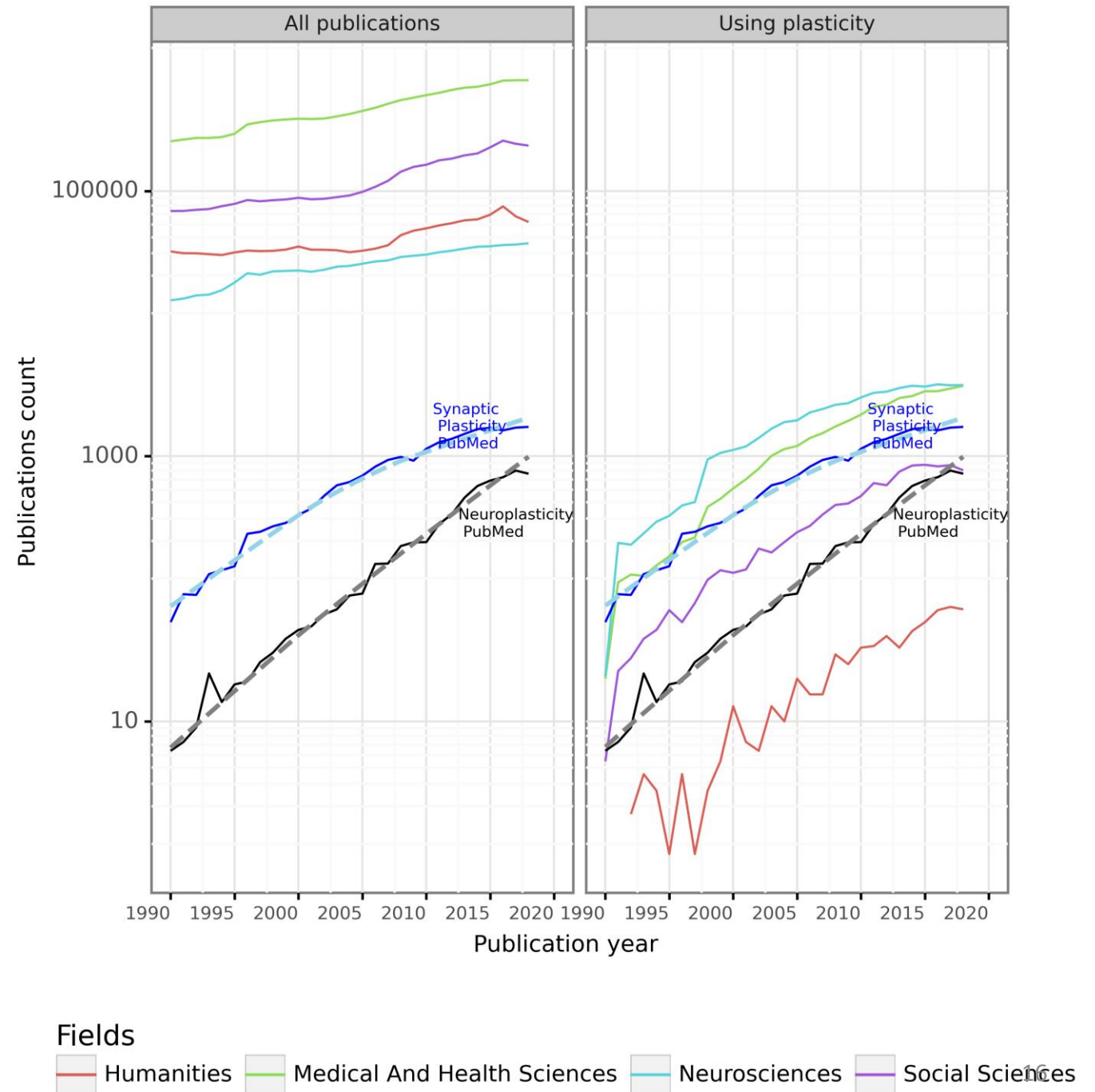
We exclude natural sciences or engineering and technology as they use this term, but their intended meaning is completely different

The increased usage of plasticity in all fields of science is evident, however the trend observed in the neurosciences shows a sharper increase



# The growth of the fields and the use of plasticity

- Neurosciences and plasticity have coevolved
- Yearly trend of article and review publications in selected fields as shown in Figure 1 (labeled as “Using plasticity”) versus the trend without limiting to publications using “plasticity” (labeled as “All publications”).
- This Figure provides a baseline for comparison to put the growth presented in Figure 1 into these fields’ context.
- We observe that the rate of growth in publications not using plasticity are rather stable and the growth rate observed among publications using this term (in neurosciences and other fields) is drastically sharper.





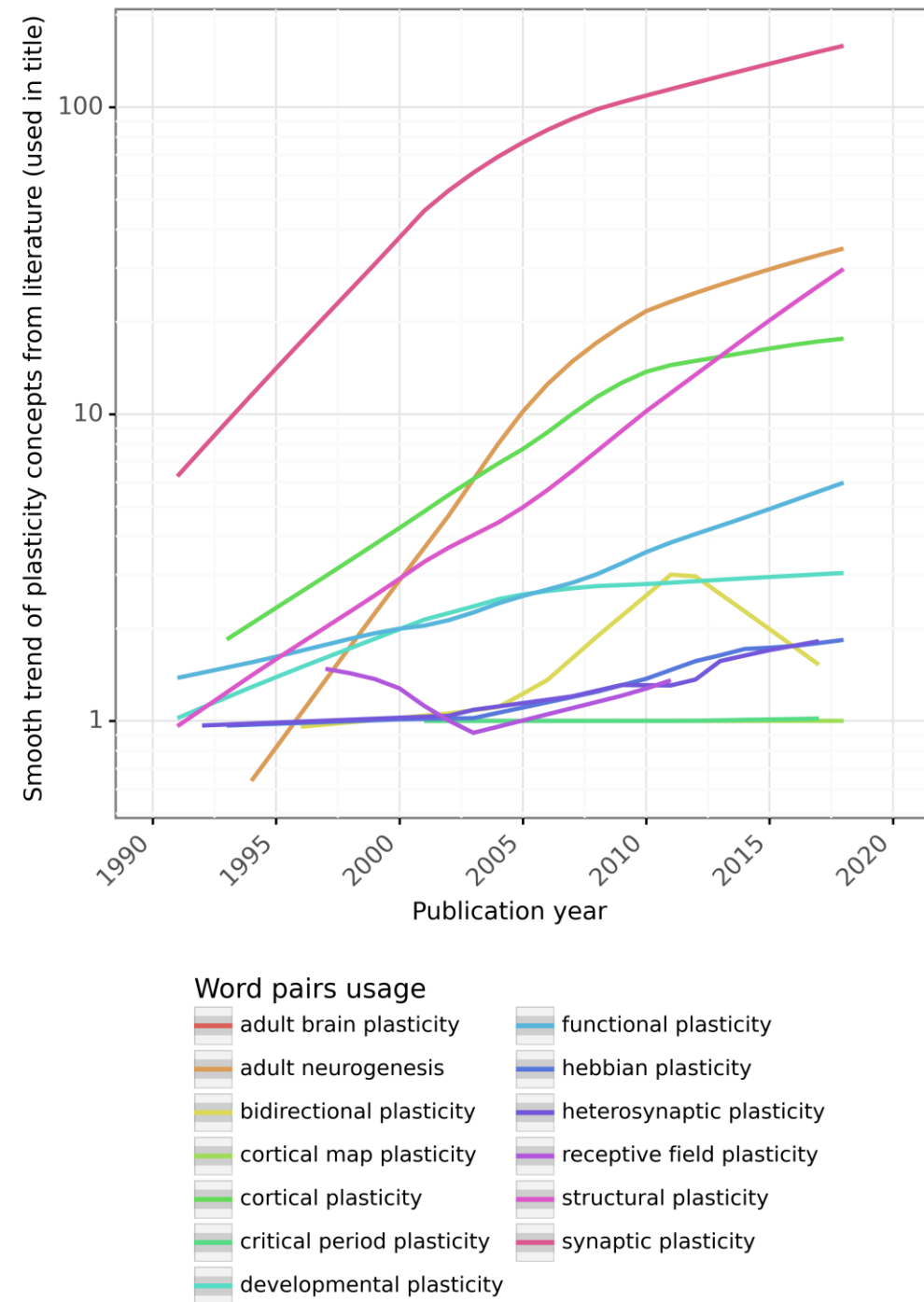
# Plasticity as a concept in the neurosciences, a word pair analysis

Keyword combinations (extracted from literature and previous systematic reviews) and temporal use of them in the **title** of scientific publications in the neurosciences (based on WOS subject categories)

Synaptic plasticity, adult neurogenesis and cortical plasticity have the highest count of publications

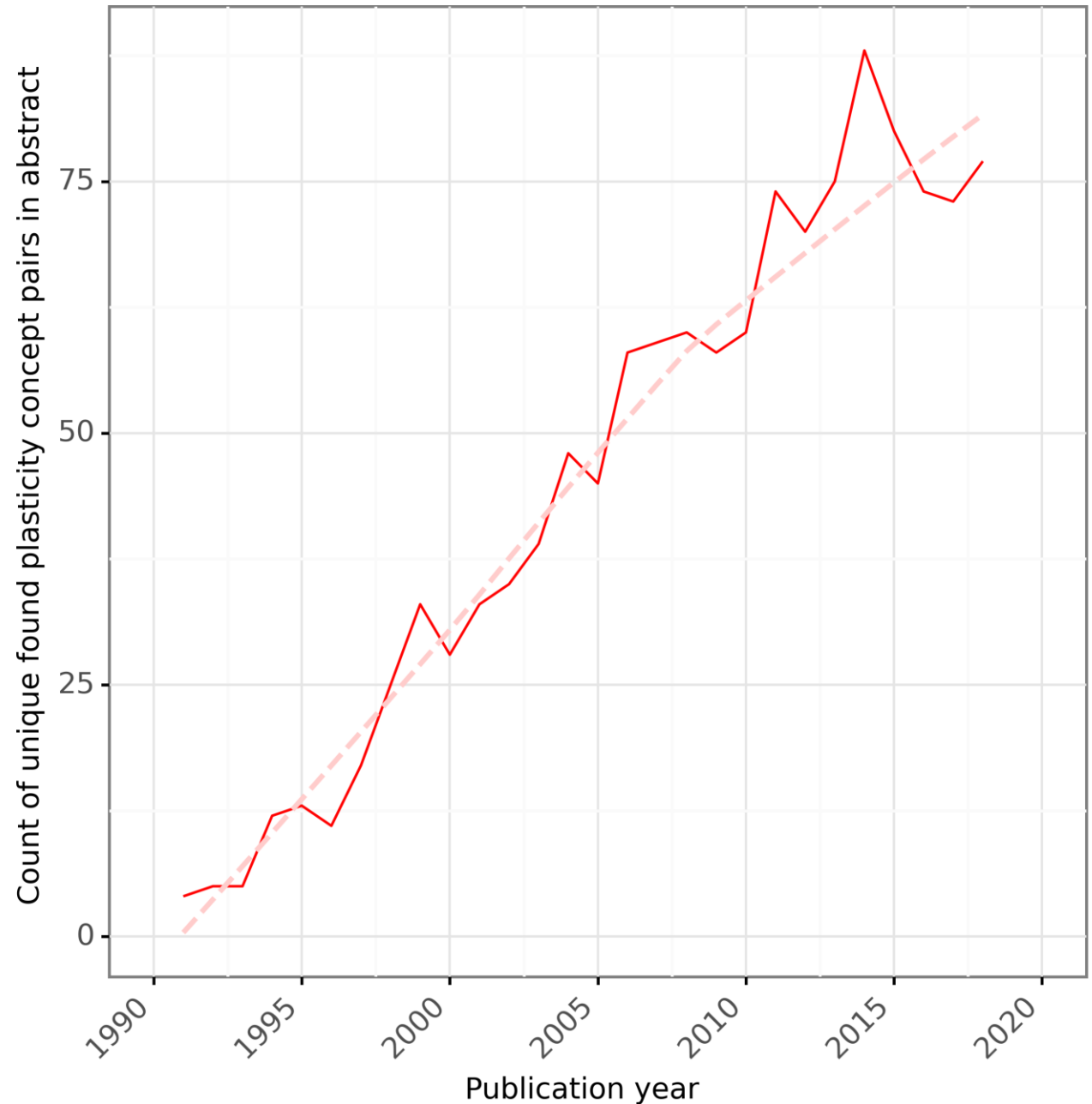
Synaptic plasticity had the earliest increase of use

Structural plasticity, adult neurogenesis, and cortical plasticity show an increase that starts in later years than synaptic plasticity



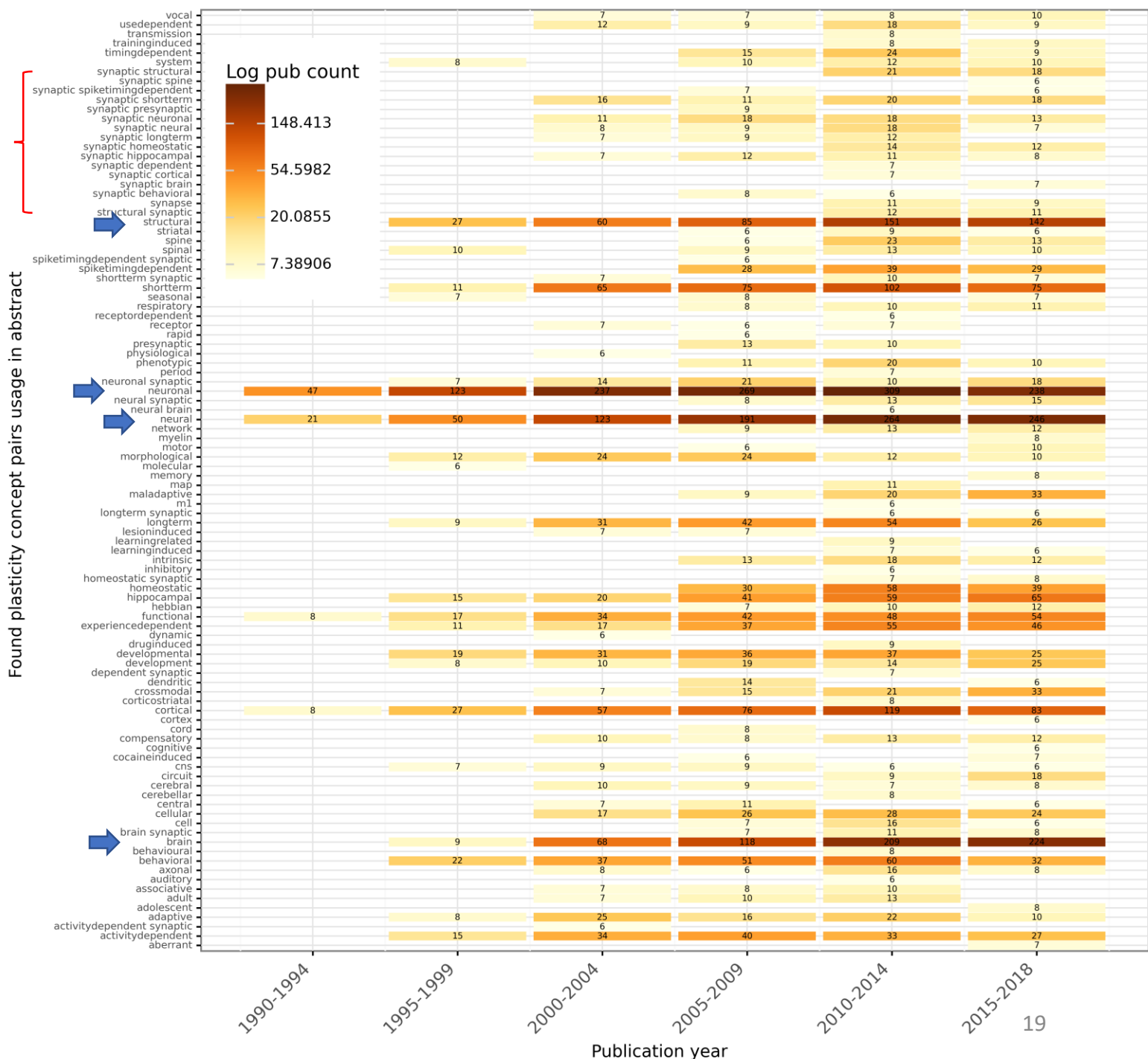
# Plasticity as a concept in the neurosciences, a word pair analysis

- Plasticity is an anchor concept
- Trend of unique word (or words in case multiple words are used in the same abstract) used before plasticity
- Red line shows the count of unique words per year and smooth trend based on mean is shown on light red dashed line.
- The count of words pairing with plasticity has been constantly on the rise since recent years. Year 2014 has the highest count of unique words (88) paired with plasticity.



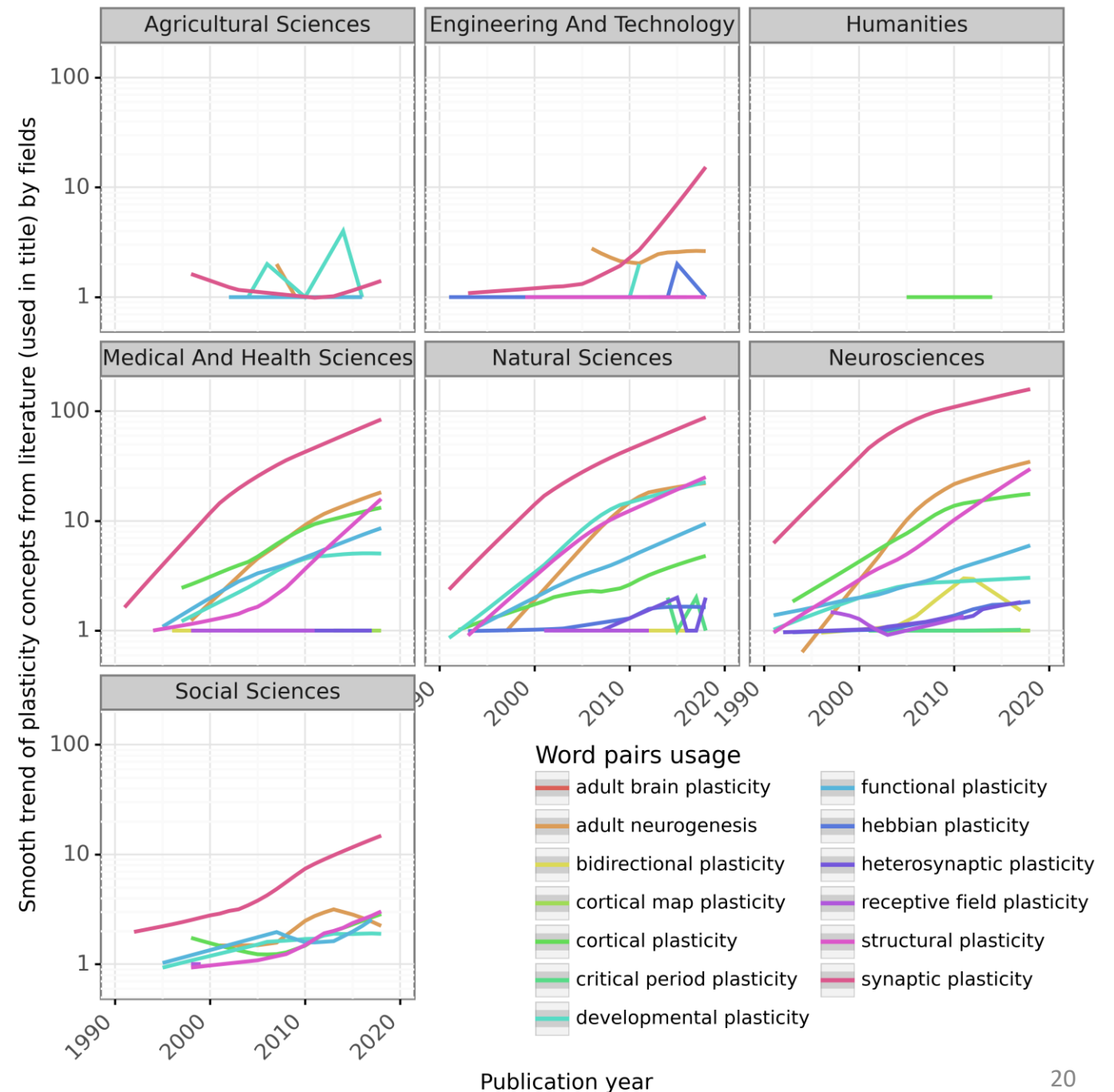
# Plasticity as a concept in the neurosciences, a word pair analysis

- Synaptic plasticity is the central anchor concept
- Found words used at least 5 times before plasticity as an anchor word and temporal use of them in the **abstract** of scientific publications in the neurosciences (based on WOS subject categories)
- If two words are used before plasticity in the same abstract, they are presented together on the Figure. We exclude “synaptic” as it dominates the visualization (see **red curly bracket for some**) and it is covered before
- *Neural, neuronal, structural, and brain* are the most frequently used words before plasticity (**blue arrows**)



# Plasticity's use in other fields

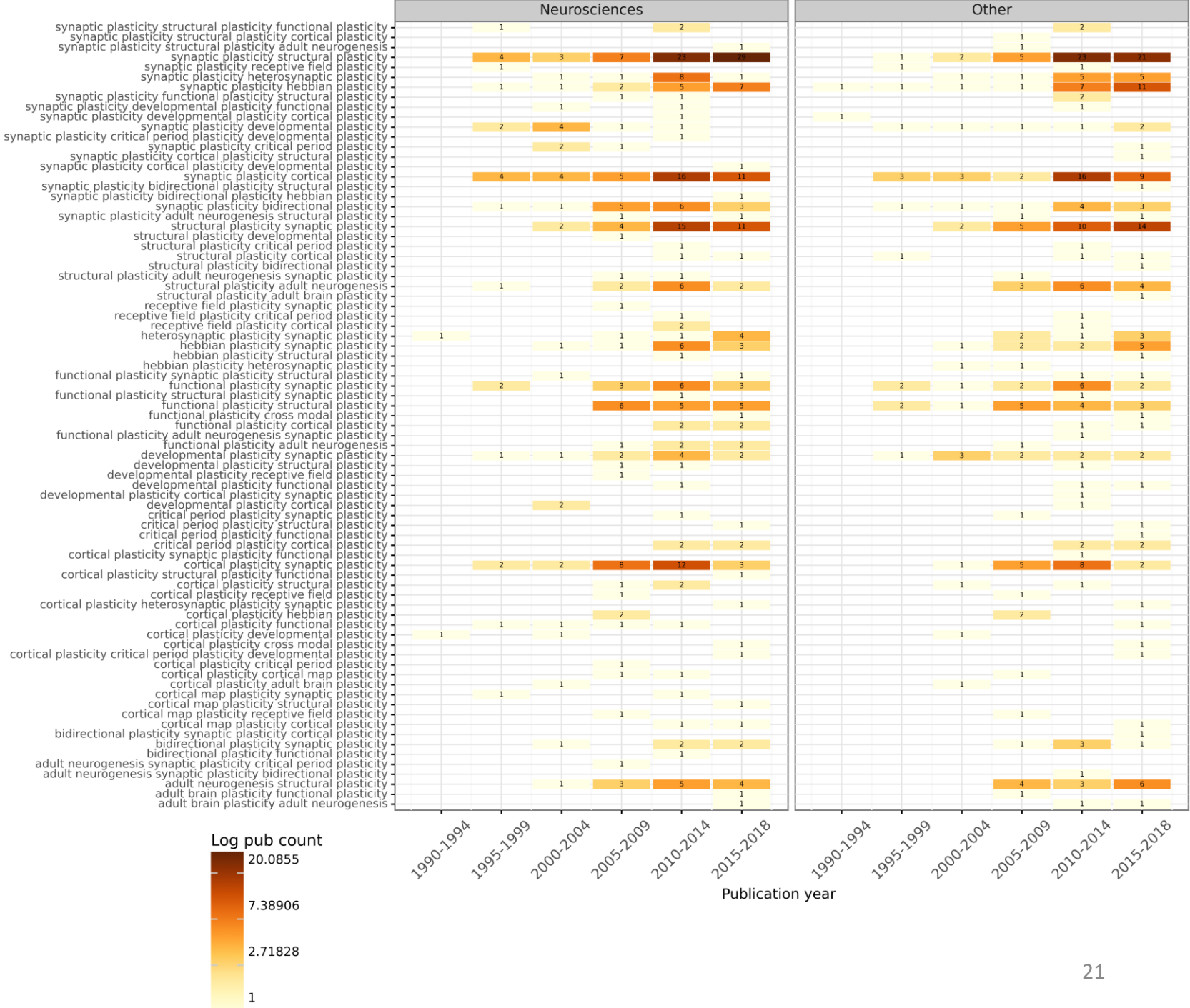
- Different conceptualizations of plasticity are differently successful
- Keyword combinations (extracted from literature and previous systematic reviews) and temporal use of them in the **title** of scientific publications in all scientific fields.
- Specific keyword combinations are borrowed from the neurosciences and used in all other fields in subsequent years, e.g., developmental plasticity and cortical plasticity that are used even in humanities (in only few publications).
- See the more recent prevalence of synaptic plasticity in engineering and technology relative to neurosciences or natural sciences.
- On the other hand, *Synaptic plasticity* and *adult neurogenesis* are the most frequently (higher counts printed inside cells) and persistently (continued blocks) used in almost all fields (excluding humanities).



# Plasticity's use in other fields

- Synaptic plasticity is the central anchor concept
- *Multiple* keyword combinations (extracted from literature and previous systematic reviews) and temporal use of them in the **abstract** of scientific publications in neurosciences versus all other scientific fields.
- Here we exclude one keyword combination and intend to highlight the *multiple* combinations used in the abstracts.
- This enables us to show when a certain combination is introduced in neurosciences and taken up in a later year by other fields of science (e.g., see *synaptic plasticity structural plasticity* where this combination is on the increase in other fields around 2010, but in neurosciences, it has already shown its increase. Similarly, see *synaptic plasticity cortical plasticity*).

Plasticity concepts from literature (used in abstract)



# Limitations

- Our corpus construction strategies are focused only on “plasticity”
- If there are other concepts with the similar permeability as plasticity in neurosciences, we have missed out on them
- Exact word search which is prevalent in literature is prone to typos or wording (word choices)
- **[Is this a better title?] Tracing plasticity: the analysis of a conceptual transformation of the neurosciences through bibliometrics (1965-2020)**

# Conclusions and Points for discussion

- The neurosciences and plasticity have co-evolved during the past sixty years
- **Synaptic plasticity** is the first conceptualisation of plasticity in an empirical context and the most successful concept till the present.
- Synaptic plasticity serves as a **reference concept** for other plasticity concepts.
- The bibliometric analysis of plasticity shows the changes in the use of the different plasticity-concepts and the ongoing diversification of the concept. It captures the transformation of the neurosciences which begins in the 1980s and continues into the present.

# Thank you!

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